Question 1:

The missing data pattern of X is Arbitary Pattern.
The missing data mechanism is "Missing Completely At Random (MCAR)".

Question 2:

Using mean imputation means finding the mean for that feature or column & replace the missing values with mean value.

For x_1 the mean is $= \frac{1+3+4+9+10+11+12+14+17}{9}$

$= 9$

So the missing value will be 9.

Question 3:

Based on linear Regression theory we can write:-

$$X_2 = \beta_0 + \beta_1 X_1$$

Putting all the values of X_2 & X_1 we will get ~~10 equations~~ 8 equations. & there is only 2 unknown variable (β_0 & β_1).

But the easiest way to find the co-efficient is using the matrix formula. which is:

$$B = (X^T X)^{-1} X^T y$$

where,

$$B = \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix} \quad X = \begin{bmatrix} 1 & X_1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 3 \\ 1 & 4 \\ 1 & 9 \\ 1 & 9 \\ 1 & 11 \\ 1 & 12 \\ 1 & 14 \\ 1 & 17 \end{bmatrix}$$

$$y = X_2 = \begin{bmatrix} 32 \\ 27 \\ 24 \\ 21 \\ 17 \\ 10 \\ 6 \\ 4 \\ 1 \end{bmatrix}$$

$$X^T X = \begin{bmatrix} 9 & 80 \\ 80 & 938 \end{bmatrix}$$

$$X^T y = \begin{bmatrix} 142 \\ 806 \end{bmatrix}$$

$$(X^T X)^{-1} = \frac{1}{2042} \begin{bmatrix} 938 & -80 \\ -80 & 9 \end{bmatrix}$$

$$\therefore B = \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix} = \frac{1}{2042} \begin{bmatrix} 938 & -80 \\ 80 & 9 \end{bmatrix} \begin{bmatrix} 142 \\ 806 \end{bmatrix}$$

$$= \frac{1}{2042}\begin{bmatrix} 133196 & -\cdot 64486 \\ -11360 & +\cdot 7254 \end{bmatrix}$$

$$= \begin{bmatrix} 33\cdot 6513 \\ -2\cdot 011 \end{bmatrix}$$

$$X_2 = \beta_0 + \beta_1 x_4$$

$$= 33\cdot 6513 + (-2\cdot 011 \times 10)$$

$$= 13\cdot 544$$

So the missing element is 13·544.

Question 4:

$$\text{NRMS error} = \frac{\|X_{estimate} - X_{original}\|_F}{\|X_{original}\|_F}$$

As all the other values are same expect the 2 missing values. The numerator is

$$= \sqrt{(9-7)^2 + (12.86-16)^2}$$

$$= 3.723$$

The Denominator = Forbenius norm of the original data

$$= \sqrt{1^2+32^2+3^2+27^2+4^2+24^2+7^2+21^2 + 19^2+17^2+10^2+16^2+11^2+10^2+12^2+6^2 + 14^2+4^2+17^2+1^2}$$

$$= \sqrt{4474}$$

$$= 66.888$$

So, NRMS error $= \frac{3.723}{66.888} = 0.056$

Question 5:

NRMS error gives us the relative error. Lower value means smaller error compared to the scale of the original data.

The error I got 0.056 is less than given value of 0.07. So this means the value I imputed is more accurate than the given NRMS.

Question 6:

Equal width binning divides the data into K intervals of equal size (in this case 3).

Min value = 13

Max value = 70

interval boundaries are: 13+3, 13+(2×3), 13+(3×3)... & so on.

Following is the table:

Interval	Values	Total no of values	Bin Mean
[13-16)	13, 15	2	14
[16-19)	16, 16	2	16
[19-22)	19, 20, 20, 21	4	20
[22-25)	22, 22	2	22
[25-28)	25, 25, 25, 25	4	25
[28-31)	30	1	30
[31-34)	33, 33	2	33
[34-37)	35, 35, 35, 35, 36	5	35.2
[37-40)	•	0	0
[40-43)	40	1	40
[43-46)	45	1	45

interval	Values	Total no of values	Bin Mean
[46-49)	46	1	46
[49-52)		0	0
[52-55)	52	1	52
[55-58)		0	0
[58-61)		0	0
[61-63)		0	0
[63-67)		0	0
[67-70)		0	0
[70-73)	70	1	70

The smoothed dataset is:

14, 14, 16, 16, 20, 20, 20, 20, 22, 22, 25, 25, 25, 25,
30, 33, 33, 35.2, 35.2, 35.2, 35.2, 40, 45,
46, 52, 70.

The best way to find outliers is to use visualization. Different kind of plots can be drawn using the data to find the outliers. For example histogram, box plot, clustering etc.

But for this dataset Box plot can be used to easily find the outliers. Also during the smoothing process it was seen that there was a huge gap between the last interval & the ~~interv~~ second last interval.

However using Boxplot,

~~Num~~ $n = 27$ (total no of values)

~~First Quartile =~~

Middle position = Q_2 The 14th value = 25

first Quartile = $Q_1 = (27+1) \times 0.25 = 7$th value

$= 20$

third Quartile = Q_3 = ~~27×0.75~~ $(27+1) \times 0.25$

$= 21$ ~~5~~ value

$IQR = Q_3 - Q_1 = 15$ $= 35$

Upper Fence = $Q_3 + (1.5 \times IQR)$

= $35 + (1.5 + 15)$

= 57.5

Lower Fence = $Q_1 - (1.5 \times IQR)$

= $20 - 22.5$

= -2.5

There is only 1 value outside the fence. That is 70.

~~Question~~

Smoothing:

In step 1 one of the Binning techniques were used. There techniques that can be used are below:-

(i) Binning with equal depth:

(i) Bin means: The values in the bins will be replaced by means

(ii) Bin median: The values will be replaced by medians.

(iii) Vo

(2) Regression: This technique will fit a regression model to the dataset & replace original values with predicted values from the model.

(3) Clustering: Using this technique similar data points will be grouped into clusters. Then the values of the cluster can be replaced using cluster centroid/mean.

(4) Combined Computer & human inspections

(5) Other smoothing techniques:

(i) Linear smoother
(ii) Moving average
(iii) Kalman filter
(iv) Kernel Smoother
(v) Smoothing spline

Question 7:

After sorting the data we get:

X_i	Class
4.6	1
4.7	1
4.9	1
5.1	1
5.5	2
5.8	3
6.3	3
6.4	2
6.9	2
7.0	2
7.1	3

total instances = 12

class distribution:

class 1 = 4

class 2 = 4

class 3 = 4

Initial Entropy we get before any split;

$$E(S) = -\sum p_i \log_2(p_i)$$

$$E(S) = -\left[\frac{4}{12}\log_2\left(\frac{4}{12}\right) + \frac{4}{12}\log_2\left(\frac{4}{12}\right) + \frac{4}{12}\log_2\left(\frac{4}{12}\right)\right]$$

$$= 1.585$$

Splits can occur between values of x_i where class label changes.

for example (i) between 5.1 & 5.5, at 5.3

(ii) " 5.5 & 5.8 at 5.65

(iii) " 6.3 & 6.9 at 6.35.

(iv) " 7.0 & 7.1 at 7.05

If we take the first one & calculate the information gain.

- First partition $(x_1 \leq 5.3)$.

  total instances = 4 (all class 1)

  $E(P_i) = 0$

Second

- Partition P2 (x_4 Y 5·3):

total instance = 8 (Class 2 = 4, Class 3 = 4)

Entropy, $E(P_2) = -\left[\frac{4}{8}\log_2\left(\frac{4}{8}\right) + \frac{4}{8}\log_2\left(\frac{4}{8}\right)\right]$

$= 1$

Combined Entropy,

$E(P_1, P_2) = \frac{4}{12}(0) + \frac{8}{12}(1.0) = 0.667$

∴ Information Gain = 1.585 - 0.667

= 0.918

if we take the second example then:

Partition P_1 ($x_4 \leq 5.65$):

total instances = 5 (class 1 = 4, class 2 = 1)

Entropy, $E(P_1) = -\left[\frac{4}{5}\log_2\frac{4}{5} + \frac{1}{5}\log_2\frac{1}{5}\right]$

= 0.722

Partition P_2 $(X_1 > 5.65)$:

total instances = 7 (class 2 = 3, class 3 = 4

Entropy $E(P_2) = -\left[0 + \frac{3}{7}\log_2\frac{3}{7} + \frac{4}{7}\log_2\frac{4}{7}\right]$

$= 0.985$

Combined Entropy

$E(P_1, P_2) = \frac{5}{12} \times (0.722) + \frac{7}{12}(0.985)$

$= 0.876$

$\therefore$ Information gain $= 1.585 - 0.876$

$= 0.709$

So the information gain is less.

For all the other examples it will be less than 0.0.

So the best split is ~~$X_1 < 5.3$~~

~~$X_1 > 5$~~ at $X_1 = 5.3$